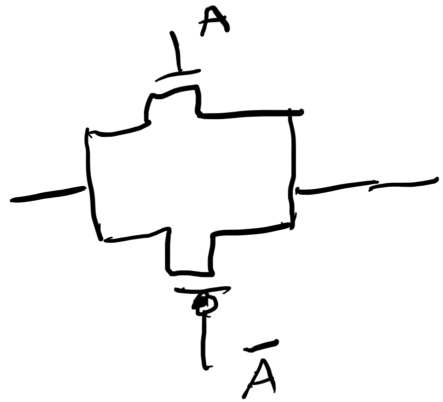
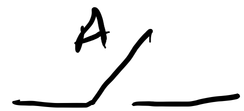
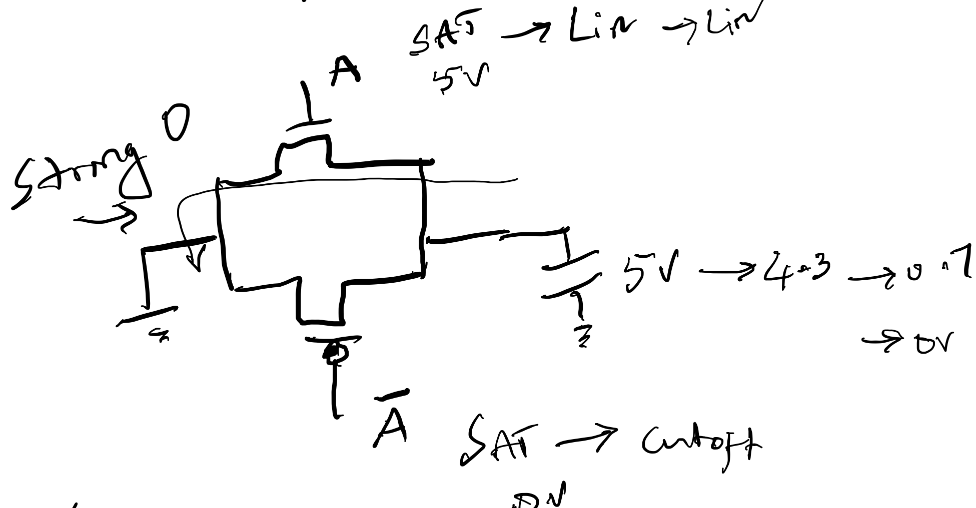
Transmission Gate (T-gate)



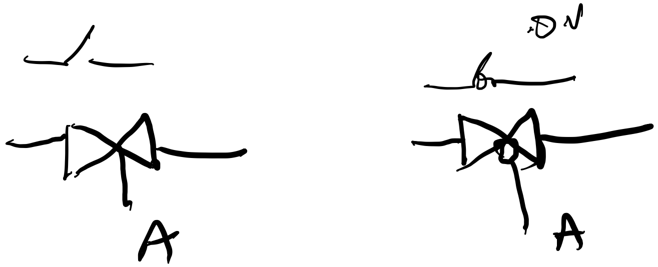


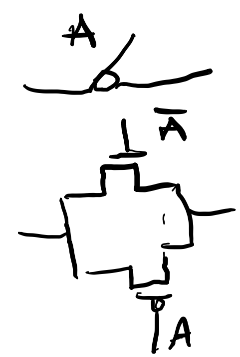
A = 0 T-gate is open

A = 1 -- closed



Good conductor of both "0" and "1"

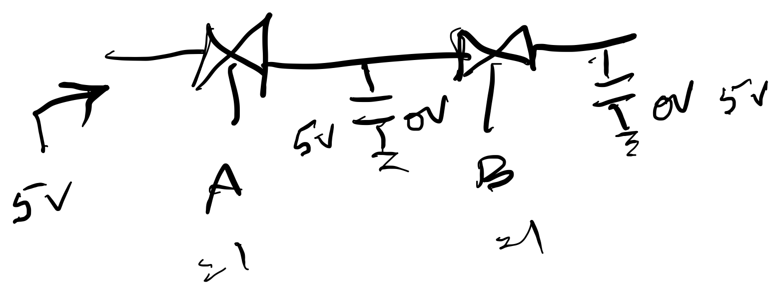


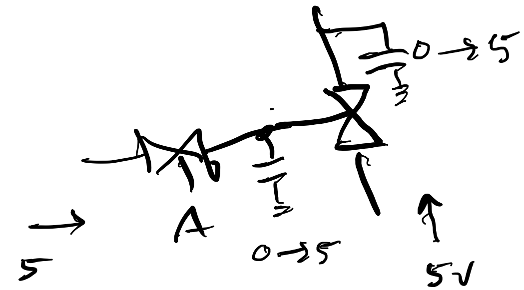


A = 1 Switch closed

A = 0 Switch open

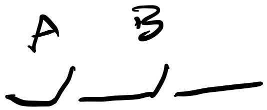
Transmission Gate – Transmission Quality

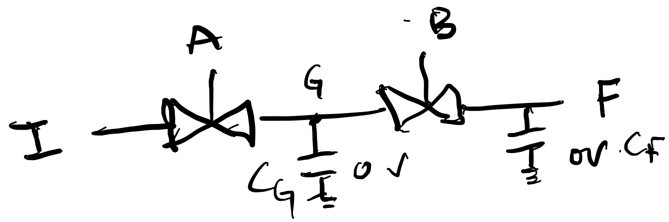




Bidirectionality and Charge Sharing

Redistribution





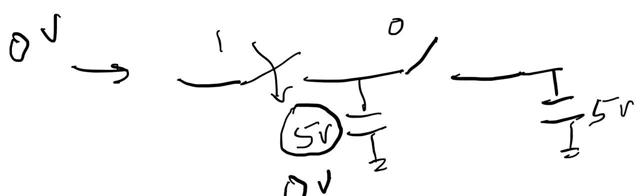
1) $A = 1, \; B = 1, \; I = 5v$

$C_G = 5v, \quad C_F = 5v$

$F = \text{"}1\text{"}$

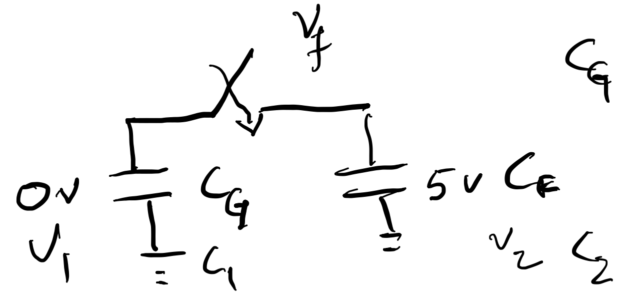
2) $A = 1, \; B = 0$

$C_G = 5v, \quad C_F = 5v$ ✓



3) $A = 0, \; B = 1, \; I = 0$

$C_G = 2.5 = C_F$



$C_G = C_F = C$

fF

$$V_1 \cdot C_1 + V_2 C_2 = V_f (C_1 + C_2)$$

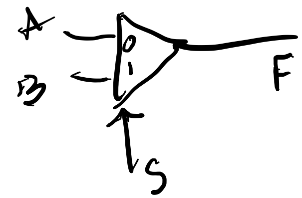
$$V_f = \frac{V_1 C_1 + V_2 C_2}{C_1 + C_2}$$

$V_1 = 0, \quad V_2 = 5, \quad C_1 = C_2$

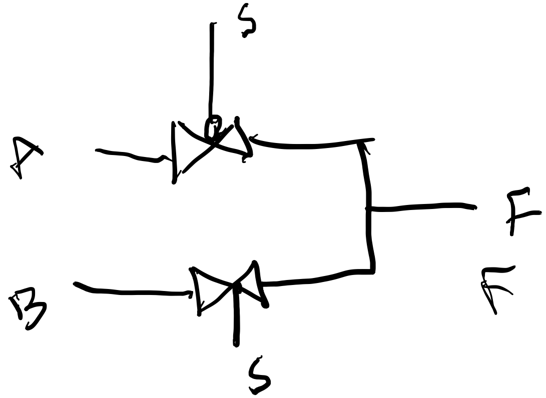
$V_f = 2.5v$

Steering Logic

Multiplexor



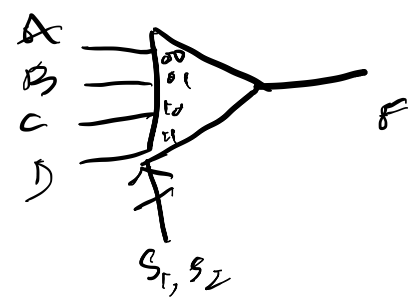
F = A when S=0
= B " S=1



F is never left floating

A, B are both strong signals

Practice Problem



Simple Memory: Charge Sharing

Reading Q

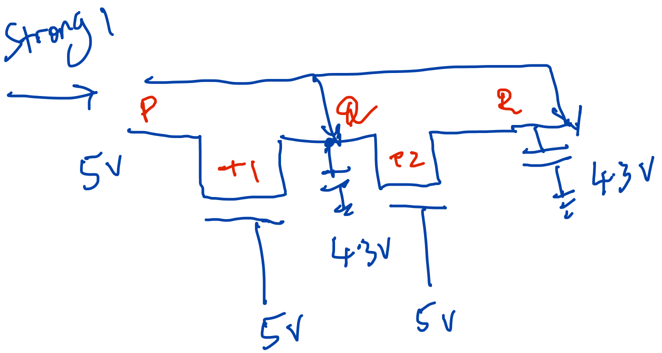
may discharge

V_Q

Quiz: CKT-1

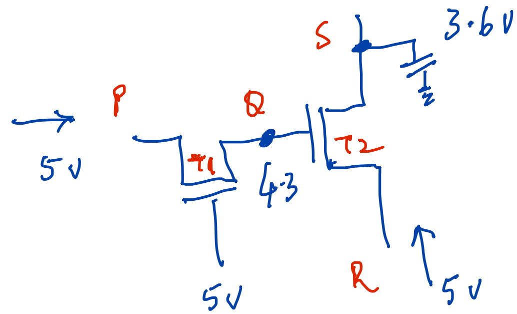
Pass Transistor Logic – Transmission Quality

Q1



Determine the SOURCE and DRAIN nodes for all transistors just before conduction stopped.

Q2

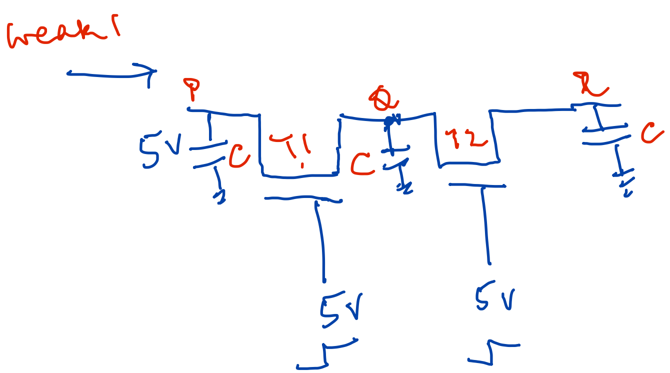


Determine the SOURCE and DRAIN nodes for all transistors just before conduction stopped.

Quiz CQI-2

Pass Transistor Logic – Transmission Quality

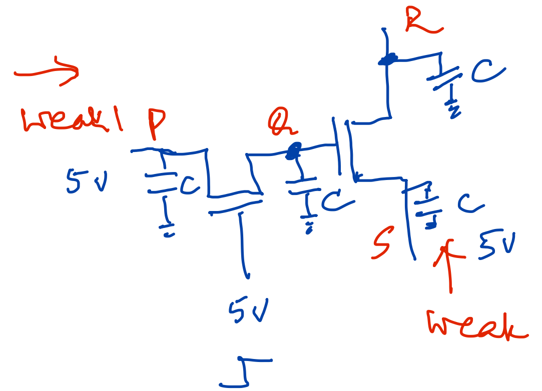
Q1



Assume $V_Q = V_R = 0$ initially.

What are the final values of V_Q and V_R.

Q2

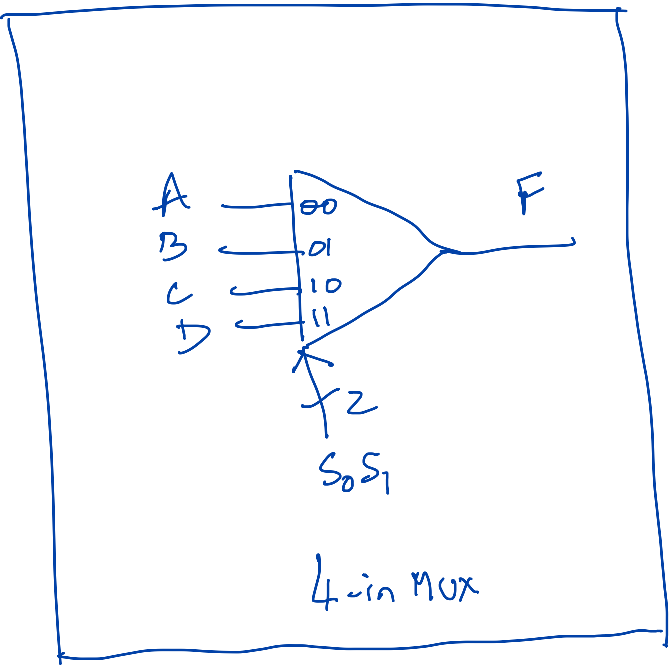


Assume $V_Q = V_R = 0$ initially.

What are the final values of V_Q and V_R.

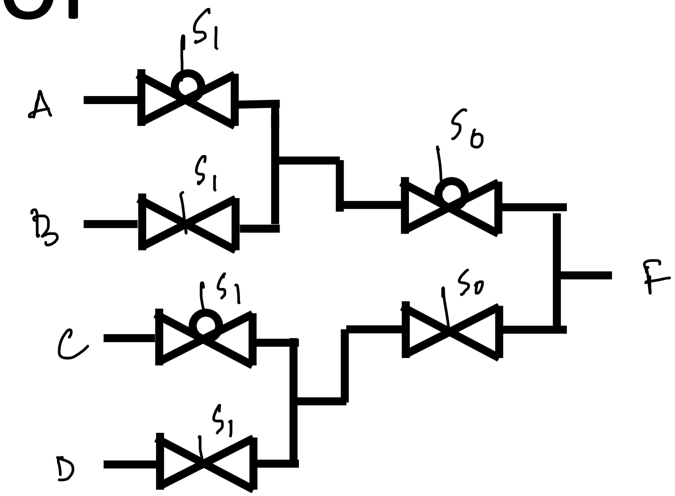
Quiz CEF_3

Multiplexor

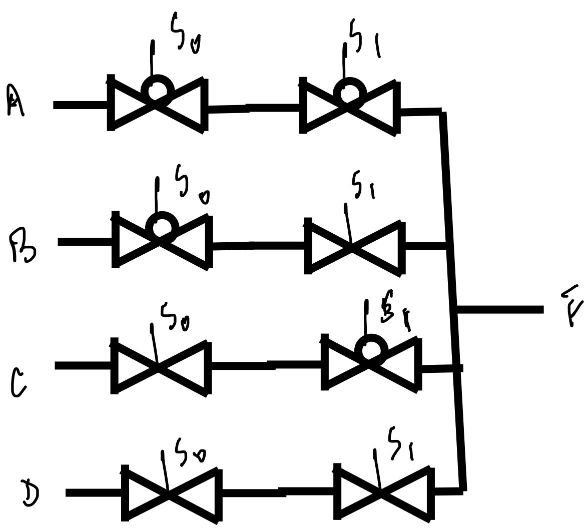


Which of these circuits is correct?

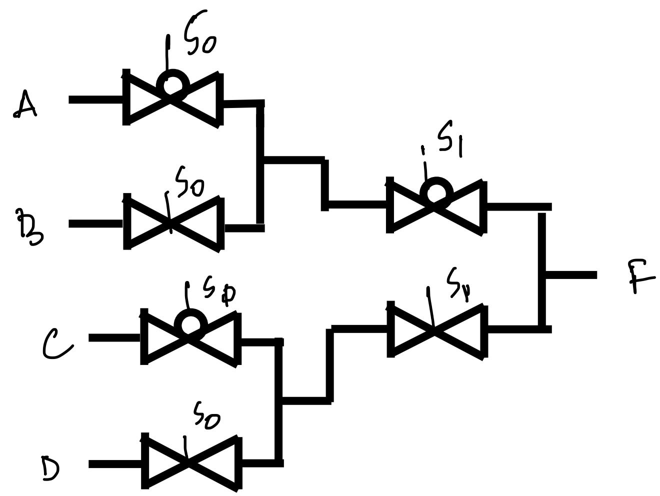
I



II

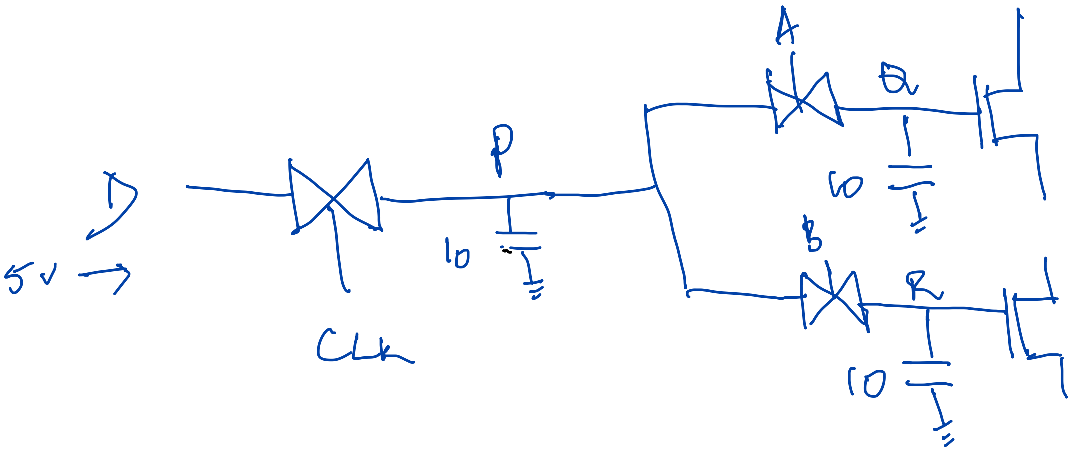


III



Quiz: C45-4

Simple Memory: Charge Sharing



All Caps are 10 fF

Assume D is a strong node. Initially $V_P = V_Q = V_R = 0V$

Consider the following sequence:

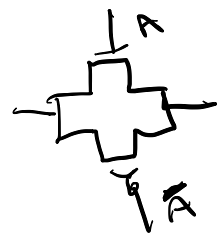
1) Memory Write: CLK = 1, A = 0, B = 0

2) Memory Read: CLK = 0, A = 1, B = 1

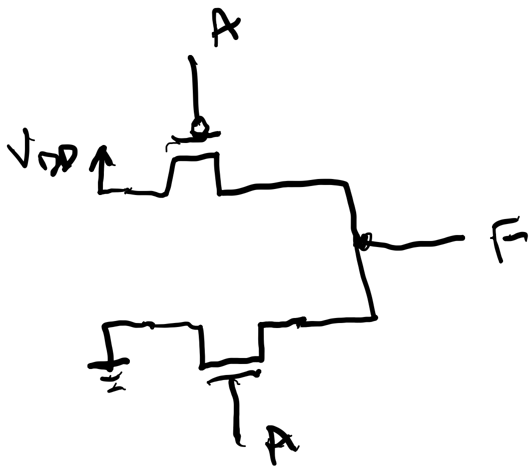
Determine the final values of V_P, V_Q, V_R

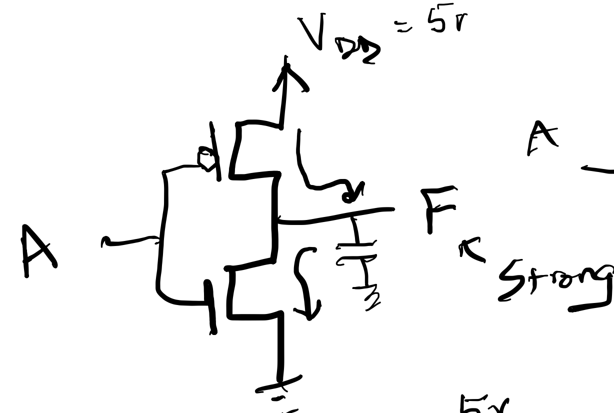
Issues With T-Gate

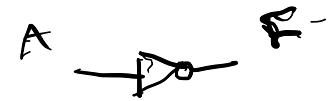
- Transistors – doubled
- Needs inverter.
- Routing
- Charge sharing
- Lack of Noise Immunity



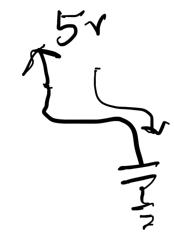
Fully Complemented Static CMOS Inverter







Static

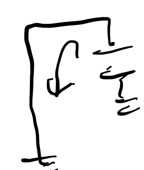


F is never floating

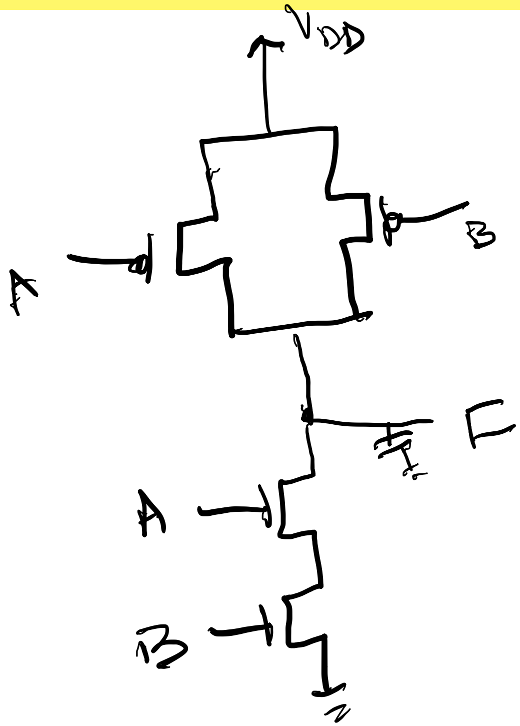
A	F
0 v	5 V
5 V	0 v

A = 0v

A = 5v

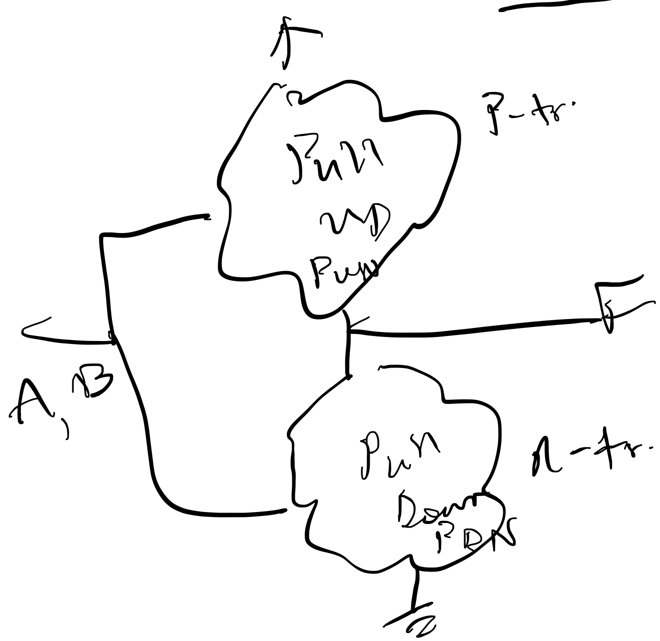


Fully Complemented Static CMOS NAND



A	B	F
0	0	1 (VDD)
0	1	1
1	0	1
1	1	0

NAND(A,B)



PUN and PDN are functional complements of each other.

Fully Complemented Static CMOS NOR

Practice